Multi-Node Multi-Band Cooperative Integrated Sensing and Communications: State-of-the-Art, Challenges and Opportunities

Haojin Li, Kaiqian Qu, Chen Sun, Shuo Wang, Xiaoxue Wang, Yujie Zhang, Basuki Priyanto, and Haijun Zhang

Abstract

Integrated sensing and communication (ISAC), a pivotal advancement in 6G networks, can increase spectrum utilization and support innovative applications such as digital twins and smart cities. Nonetheless, existing single-node ISAC networks face challenges such as restricted coverage, limited sensing capabilities, and the inability to meet the stringent quality of service (QoS) requirements for combined sensing and communication. Multi-node multi-band cooperative ISAC (MNMB-ISAC) networks have emerged as a promising strategy to overcome these challenges. In this study, we introduce a user-demand-driven MNMB-ISAC framework that exploits the advantages of ISAC, multi-node cooperative ISAC, and multi-band cooperative ISAC. This network architecture is designed to provide specialized services for a variety of communications or sensing requirements, substantially improving the network performance through cooperative resource allocation strategies. The empirical performance evaluations confirm the efficacy of the MNMB-ISAC approach in ensuring high QoS, guaranteeing mobility services, and improving edge coverage. Furthermore, we examine the challenges encountered by MNMB-ISAC and prospective research avenues. This work is expected to stimulate further breakthroughs in the domain of cooperative ISAC.

Introduction

As 6G network research advances, cutting-edge applications, such as autonomous driving, smart cities, and digital twins, have imposed stringent requirements on the sensing capabilities of wireless networks. Integrated sensing and communication (ISAC) has emerged as a pivotal technology underpinning 6G networks, attracting substantial interest from both academia and industry [1]. By sharing hardware, spectra, and even signals, 6G networks will possess both communication and sensing capabilities. Additionally, by leveraging the strong networking capabilities of communication systems, ISAC-enabled mobile networks can achieve comprehensive environmental sensing, providing technical support for environment-aware applications [2].

ISAC has been extensively studied in both the communication and radar domains, with numerous survey papers providing an in-depth overview of the fundamental concepts and existing integration methods [1–7]. However, existing research (Table 1) often centers on single-node ISAC scenarios, which limits sensing coverage and accuracy. Considering the inherent multi-band base station (BS) distribution in contemporary communication networks, leveraging cooperation among different nodes and frequency bands is anticipated to overcome the constraints associated with single-node ISAC. Furthermore, some studies have explored multi-node cooperation or multi-band cooperation, such as coordinated multipoint (CoMP) transmission [3] and dual-mode BS [8]. However, a comprehensive ISAC strategy involving multi-band and multi-node cooperation has not been thoroughly discussed. Its significant advantages can be summarized as follows:

- First, multi-node cooperation can significantly expand coverage, improving both mobility service assurance and edge coverage. This approach creates additional echo paths and wider sensing angles for sensing applications. Multi-node cooperation can also reduce interference from other nodes and enhance the strength of useful signals.
- Second, multi-band cooperation can fully leverage the advantages of each frequency band. The sub-6 GHz frequency band, though limited in bandwidth, has low path loss, making it suitable for macro BS, providing wide communication coverage and low-precision long-distance sensing service. In contrast, the millimeter-wave frequency band, with its abundant bandwidth but higher path loss, is ideal for high-capacity communication and high-precision short-range detection in hotspot areas. Consequently, multi-band cooperation can flexibly meet diverse sensing and communication requirements.
- Finally, from an optimization perspective, multi-node and multi-band cooperative (MNMB) offers more optimization degrees of freedom (DoFs), improving both sensing and communication performance.

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Ref	ISAC level	Multi-node collaboration	Multi-band collaboration	Collaboration scenario	Optimization variables	Communication and sensing metrics
[3]	RCC	✓	×	CoMP-JP for communication	Communication transmit precoder; Radar transmit precoder	Mutual interference
[6]	DFRC	×	×	—	DFRC transmit precoder	SINR; Beampattern
[9]	DFRC	✓	×	CoMP-CS for communication; Multi-BS sensing interference mitigation.	Collaborative transmit precoder	SINR; SINR
[10]	Sensing-only	✓	✓	Macro and Micro BS cooperative sensing	No optimization variables (Collaborative sensing data processing algorithms)	RMSE
[11]	DFRC	✓	×	C-RAN based ISAC network (CoMP-JP for communication; Distributed-MIMO radar for sensing)	DFRC transmit precoder and receive filter	SINR; SINR
[12]	RCC	✓	×	C-RAN based ISAC network (Some nodes perform COMP-JP, some nodes perform Multi-node sensing)	Mode selection and power control	SINR; MASR
[13]	DFRC	✓	×	C-RAN based ISAC network (CoMP-JP for communication; Distributed-MIMO radar for sensing)	Cooperative BS cluster sizes and the transmit power	SINR; CRLB
[14]	Comm-only	×	✓	Dual-mode BS (mmWave and sub-6G)	Transmit precoder and user scheduling	SINR

TABLE 1. Summary of research on cooperation ISAC in terms of level, cooperation type, cooperation scenario, optimization variables, and metrics.

Therefore, it is necessary to explore and discuss MNMB-ISAC. On the basis of the above analysis, we provide a comprehensive overview of cooperative ISAC, propose the user-demand-driven MNMB-ISAC framework, and identify existing issues and key challenges. The next section provides a brief review of existing works on cooperative ISAC. Then, we briefly discuss the typical application scenarios for MNMB-ISAC. Following that, we present a new user-demand-based MNMB-ISAC framework and provide simulation verification of MNMB-ISAC from a resource allocation perspective. We then highlight the open issues and challenges for MNMB-ISAC. The final section concludes this work.

Overview of Cooperative ISAC

In this section, we provide the development status of ISAC, including cooperative ISAC, and provide a brief review of existing works.

Principles of ISAC

Over the past few decades, communication and radar technologies have evolved in parallel. With the rapid proliferation of connected devices, spectrum resources have become increasingly scarce, necessitating the shared use of the spectrum between communication and radar systems to mitigate this issue. The hardware similarities between communication and radar systems have facilitated the realization of ISAC technology.

ISAC enables two distinct levels of integration: radar-communication coexistence (RCC) and dual-function radar communication (DFRC). In RCC, the primary focus is the coexistence of radar and communication systems within the same frequency band, and the main objective is to eliminate mutual interference between these systems [3]. RCC necessitates information exchange between communication and radar systems, which incurs additional overhead and does not lead to mutual benefits. Conversely, DFRC uses a shared hardware system for radar sensing and communication transmission, eliminating the need for additional information exchange and allowing sensing and communications to mutually assist each other. DFRC transmitters typically emit a unified waveform that can be implemented across different resource levels. For example, [5] implemented ISAC using different orthogonal subcarriers involving the allocation of frequency and power resources, whereas [4] achieved ISAC through different time slots involving the allocation of time resources. Additionally, ISAC can be realized through space and code division methods. Fully integrated DFRC systems further advance this concept by breaking the constraint that sensing and communications must be allocated to orthogonal resource blocks. These systems achieve ISAC by optimizing a unified transmit waveform or precoding matrix, incorporating embedded power allocation optimization.

However, most of the aforementioned studies are based on single-node and single-band scenarios and do not leverage the cooperation of multi-node and multi-band in wireless networks to improve system performance. Therefore, the principles discussed here serve as the foundational understanding necessary to explore the more advanced concept of MNMB-ISAC.

Muti-Node Cooperative ISAC

Due to the complexity and variability of the environment, single-node ISAC systems often face limitations in performance, particularly in terms of coverage range and resource availability. Multi-node cooperative ISAC offers the potential to improve the sensing and communication performance in the network. There are three reasons why multi-node cooperative ISAC is considered to be a future development trend:

- For communication: Multi-node cooperation can either cooperate to provide data transmission for the same user, enhancing edge user

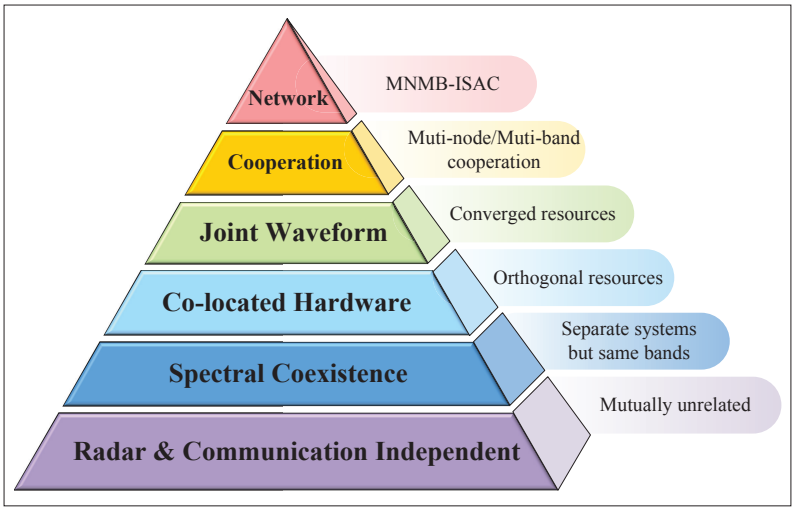


FIGURE 1. Schematic diagram of trends in ISAC development.

coverage performance, or eliminate inter-cell interference through cooperative beamforming. Multi-node cooperation can better support high mobility users by providing seamless handover and continuous service, ensuring the quality of communication for mobile users (similar advantages apply to sensing as well).

- For sensing: Multi-node cooperation forming a distributed MIMO radar can improve the sensing coverage, improve target detection probability, and increase the localization accuracy. Multi-node setups provide different observation angles, yield richer information, and thereby improve sensing capabilities. Furthermore, in multi-node cooperative sensing, there are different sensing modes: cooperative active sensing, cooperative passive sensing, and cooperative active and passive sensing.
- Multi-node cooperative resource scheduling offers more DoFs for performance optimization, thereby improving the system performance. Moreover, the inherent BS distribution in wireless networks provides hardware support for multi-node cooperative ISAC. For example, [9] proposed a perceptive mobile network architecture based on cloud radio access networks (C-RANs), which rely on BSs and cloud centers in cellular networks.

Recently, several studies have been conducted on multi-node cooperation for different ISAC systems.

Multi-Node ISAC in RCC systems: Reference [3] proposed a method for achieving multi-node cooperative communication in RCC systems and eliminates mutual interference by designing precoding for both communication and radar. The authors also studied a multi-node cooperative RCC system [10]. However, cooperative node selection was considered, optimizing BS mode selection and power control to maximize the communication signal to interference plus noise ratio (SINR) while maintaining the main-lobe to average side-lobe ratio (MASR).

Multi-Node ISAC in DFRC Systems: The authors studied the cooperative precoding design in multi-BS systems [11], aiming to eliminate communication interference between BSs, sensing interference between multiple BSs, and interference between sensing and communication systems. The aim of the study in [11] was to achieve interference management among multiple nodes without fully leveraging multi-node cooperation. The joint design of precoding and receiving filters in C-RAN networks to enhance the SINR of users and echoes was studied [12]. By considering interference from outside the cooperative cluster, the study [13] optimized the size of the cooperative cluster within the same network structure.

These discussions on multi-node cooperation highlight the potential enhancements in system performance and pave the way for the MNMB-ISAC framework proposed in this article.

Multi-Band Cooperative ISAC

The efficacy of sensing and communication systems is intrinsically tied to the utilization of frequency bands, which significantly influence their operational range and performance. Each frequency band offers distinct advantages and limitations, as discussed in the introduction. By optimally leveraging the synergy between diverse frequency bands, their complementary strengths can be utilized to enhance the overall performance of sensing and communication systems.

For Communication: Multi-band cooperation enhances coverage by addressing the high path loss and limited coverage of high-frequency bands. It also capitalizes on the abundant bandwidth of these bands to facilitate high-speed data transmission. Advanced technologies, such as carrier aggregation in LTE-A and 5G NR, enable cross-band aggregation, improving communication rates. Moreover, multi-band cooperation allows flexible resource allocation to meet diverse network demands through dynamic spectrum access.

For Sensing: Multi-band cooperation effectively supports both near and far sensing requirements. Low-frequency bands, with their extended range, are ideal for long-distance sensing, while high-frequency bands excel at precise, short-range detection. This two-stage approach, using low-frequency bands for coarse sensing and high-frequency bands for fine sensing, reduces system complexity. Additionally, integrating sensing information across bands further enhances overall system performance, including communication quality.

There are fewer studies addressing multi-band cooperative ISAC. In [8], the authors studied the joint optimization of user scheduling and precoding in dual-mode networks (millimeter-wave and sub-6G), where each user can be scheduled in both frequency bands. An algorithm for cooperative passive sensing with high and low-frequency band BSs was proposed [13], performing symbol-level fusion processing on the received signals from both high and low-frequency band BSs.

These discussions on multi-band cooperation emphasize the necessity of integrating multiple frequency bands to enhance overall system performance, leading naturally to the proposed MNMB-ISAC framework in this article.

Multi-node or multi-band studies have largely focused on specific technical aspects, such as multi-node waveform optimization, interference management, and data fusion on the receiver side, typically in the context of radar-only or communication-only applications. However, they generally lack a holistic network-level framework that

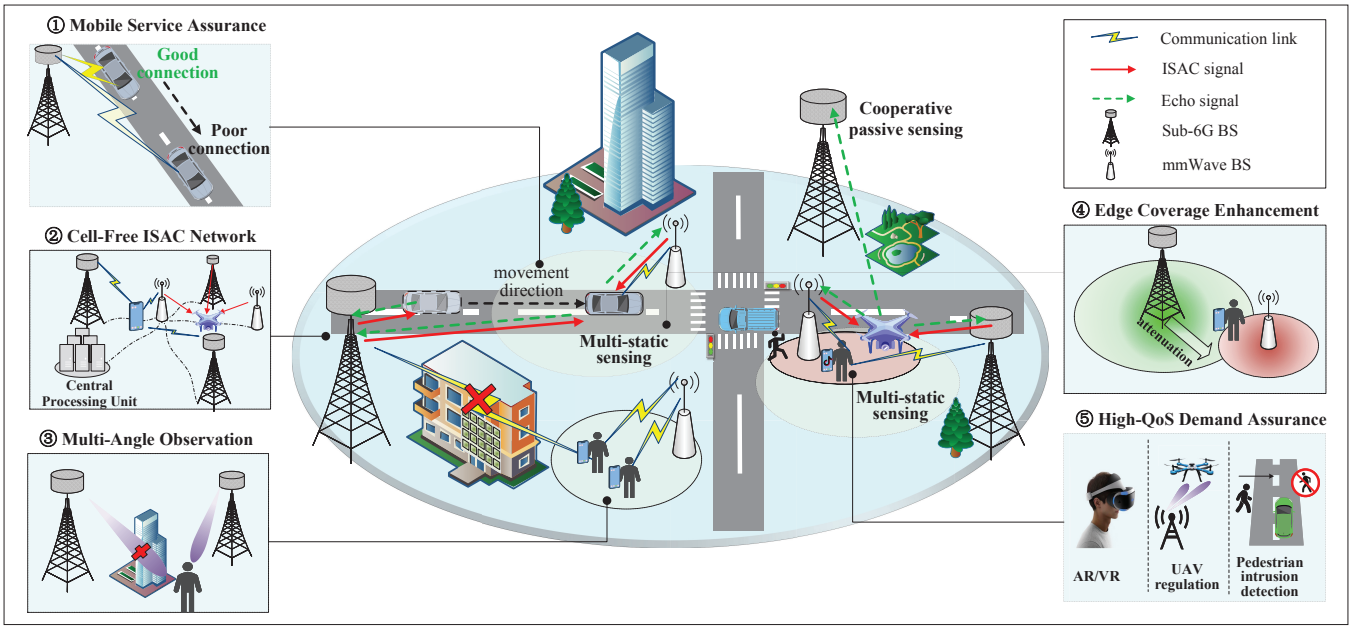


FIGURE 2. Schematic diagram of an ISAC scenario with multi-node multi-band cooperation.

addresses user-centric ISAC applications, particularly one that integrates both multi-band and multi-node features. Therefore, MNMB-ISAC represents a future development trend [14], as shown in Fig. 1. In the next section, we analyze several specific cases to demonstrate the importance of MNMB-ISAC research.

Application Case Studies

In this section, we explore typical cases of MNMB-ISAC in practical applications, as shown in Fig. 2. These cases demonstrate the advantages and potential applications of multi-node multi-band cooperation in future ISAC networks.

ISAC Mobile Service Assurance

Mobile service assurance is an important application scenario for MNMB-ISAC. In mobile networks, user positions constantly change, and ensuring continuous, high-quality service during user mobility is a significant challenge. Through multi-node cooperation, MNMB-ISAC can provide extensive coverage and greater service continuity. Multiple nodes can work cooperatively, flexibly allocating resources on the basis of user location reports and experience feedback. Additionally, multi-band cooperation can fully utilize the advantages of different frequency bands to provide optimal coverage and service for sensing and communication in various environments.

Cell-Free ISAC Network

Cell-free ISAC networks are another important application scenario. Traditional cellular networks rely on fixed BS layouts, whereas cell-free networks provide services through the cooperation of multiple distributed nodes. In cell-free networks, nodes need to dynamically adjust their roles and resource allocation to flexibly respond to different service demands and environmental changes. In this architecture, MNMB-ISAC can fully exploit its multi-node and multi-band cooperative advantages, improving the network's ISAC performance.

Multi-Angle Target Detection

Detecting small or low radar cross-section targets in complex environments is challenging due to occlusion, multipath interference, and clutter. The MNMB-ISAC framework enhances target detection by leveraging multi-node cooperation, where radar nodes observe the same target from multiple angles. This spatial diversity improves signal-to-noise ratio (SNR), compensates for occluded signals, and increases detection probability through data fusion. Multi-band cooperation complements this by combining low-frequency bands for robust long-range detection with high-frequency bands for fine resolution. By integrating data across nodes and bands, MNMB-ISAC enables precise and reliable radar sensing, supporting applications like surveillance, traffic monitoring, and autonomous driving.

ISAC Edge Coverage Enhancement

In edge coverage enhancement scenarios, traditional single-node coverage often suffers from weak signals and high interference at the network edges, whereas MNMB-ISAC can significantly improve the signal quality in network edge areas or coverage holes. Multi-node cooperation can improve ISAC signal strength in these areas through joint transmission. Moreover, multi-band cooperation can make use of macro BSs in low-frequency bands provide wide-area coverage, and micro BSs in high-frequency bands provide hotspot coverage, working together to further improve edge coverage.

ISAC High-QoS Demand Assurance

QoS assurance is a core requirement in many critical application scenarios, such as intrusion detection and augmented reality (AR). Through multi-node cooperation and dynamic resource scheduling, MNMB-ISAC can flexibly allocate resources on the basis of user demands, ensuring QoS requirements such as communication rates, detection probabilities, and localization accuracy. Cooperation across different frequency bands can enable the selection the optimal frequency band

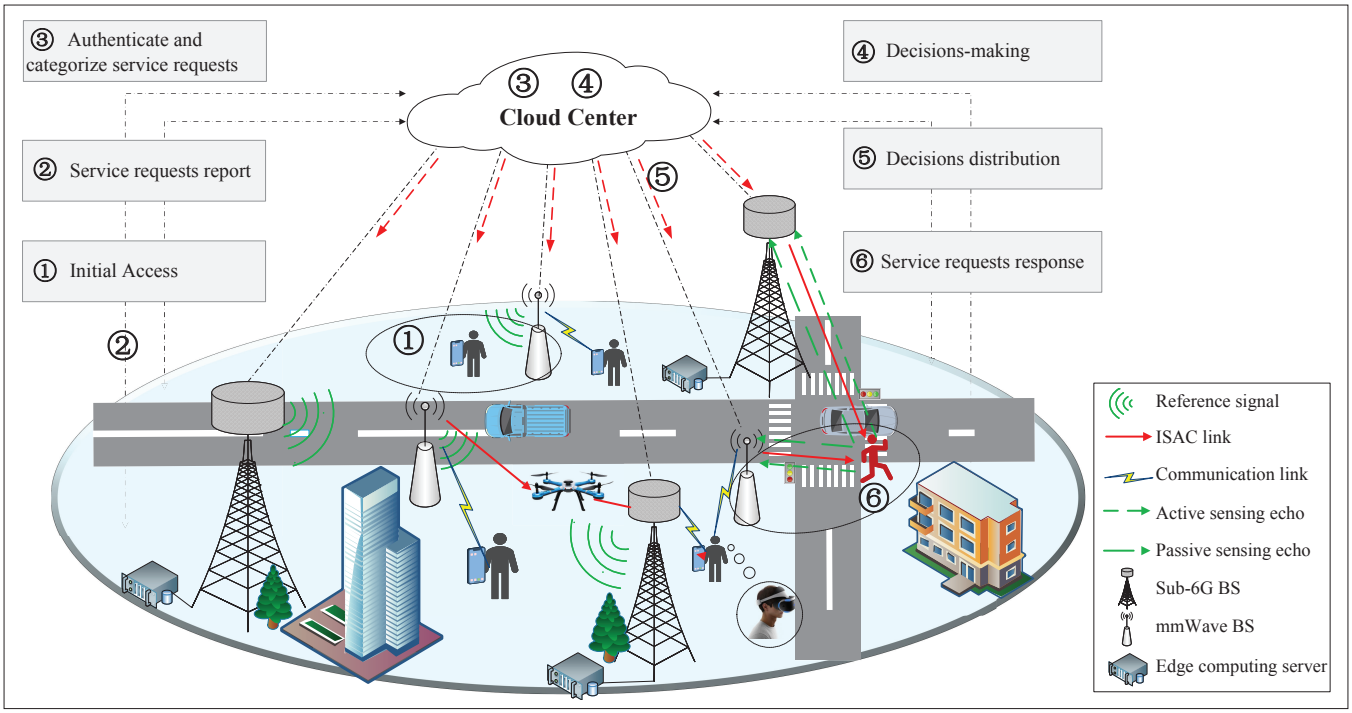


FIGURE 3. Framework of the user-demand-driven MNMB-ISAC network.

combinations on the basis of actual needs, maximizing resource utilization efficiency.

Therefore, through multi-node and multi-band cooperation, customized services can be provided on the basis of different user needs, realizing a user-demand-driven MNMB-ISAC network structure.

MNMB-ISAC: A New Cooperative Framework

In this section, we introduce the user-demand-driven MNMB-ISAC network framework and provide simulation results to verify the advantages of MNMB-ISAC.

Framework of MNMB-ISAC

As shown in Fig. 3, the proposed MNMB-ISAC network consists of high and low-band BSs, edge computing servers (ESCs), and a cloud center. The cloud center is responsible for coordinating the entire network, acting as the intelligent brain. The cloud center, which has powerful computing and storage capabilities and centrally controls all BSs in the network, is responsible for aggregating information from BSs and user equipment (UE) and processing data. By gathering global information, the cloud center makes decisions and allocates resources for the diverse services in the network, and distributes these decisions to the BSs. The basic process of the MNMB-ISAC network is as follows:

Initial Access: Prior to transmitting data via the communication network. The UE needs to select a BS for access. BSs broadcast reference signals (RSs) and the UE scans and receives RSs from various BSs. On the basis of the reference signal received power (RSRP), the UE selects a BS for access. Moreover, the UE establishes a default connection with the cloud center and reports the initial access status via the uplink.

Service Requests Report: After initial access, the UE reports service requests to the cloud center, including information such as the service type, request priority (e.g., allocation and retention priority in 3GPP standards), communication rate requirements, latency requirements, sensing identification accuracy, and sensing localization precision. Additionally, service requests can be initiated not only by the UE itself but also by BS nodes or the application layer. For example, if a BS or region needs to perform high-precision unmanned aerial vehicle (UAV) monitoring tasks, the node or regional operator also needs to report cooperative service requests to the cloud center.

Authenticate and Categorize Service Requests: Since sensing tasks may involve user privacy and security, UE requests must be authenticated, and illegal requests will be rejected. Classification involves an initial assessment of user needs. If the initially accessed BS can meet the user requirements, the original connection is maintained; otherwise, if the initial access cannot meet the demands, other nodes need to cooperate, and a deeper decision-making process is initiated.

In addition, the network dynamically adjusts user priorities during this phase based on real-time conditions. For instance, when network resources cannot satisfy all user demands, the network flexibly reassigns priorities according to factors such as service type, latency requirements, and mobility. This priority adjustment mechanism ensures that critical tasks are fulfilled even under resource-constrained conditions, maintaining the overall efficiency and fairness of the system.

Decisions Making: Based on the previous analysis and request priorities, the cloud center makes decisions according to user needs and capabilities of each BS node, including cooperation schemes, sensing modes, resource allocation and precoding.

Decisions Distribution: The cloud center distributes decisions to all BS nodes involved in cooperative ISAC via fiber links.

Service Requests Response: Based on the decisions, BSs execute the corresponding cooperative tasks, resource allocations, precoding, and so on

to complete the service response.[1] For cooperative sensing tasks, cooperative BSs report sensing data to the cloud center, which may include sensing measurement data or sensing results. The cloud center performs data fusion computations to generate the final results and returns the sensing results to the sensing service requester, completing the sensing task response. UEs and BSs periodically report service quality to the cloud center, which then performs further optimization and adjustments to ensure effective utilization of network resources.

In the aforementioned process, the computational tasks performed by the cloud center can be executed on the ESC, reducing the burden on the cloud center.

Remark 1: Our focus is on a user-driven MNMB-ISAC network framework. Within the MNMB-ISAC network, there are numerous processes, and existing research on cooperative strategies can be utilized to fill the corresponding components. For instance, the existing joint transmission/reception strategy in CoMP can be utilized to achieve multi-node cooperative transmission in the proposed network; carrier aggregation can be applied for multi-band cooperative transmission; and existing multi-band/multi-site radar data fusion techniques can be used for cooperative sensing data processing.

Performance Evaluation

To demonstrate the advantages of MNMB-ISAC, we consider a specific evaluation scenario. As shown in Fig. 4a, the sub-6G macro BS and mmWave micro BS each serve two static communication and sensing users. Additionally, the macro BS serves mobile communication, sensing, and ISAC users. The macro BS and micro BSs must meet the requirements of static users, with a communication rate of 4 Mb/s and a sensing detection probability of 90 percent. Additionally, the ISAC users have a sensing requirement with a detection probability of 95 percent, and the sensing users require localization services. The optimization objective is to maximize the communication rate of mobile users while ensuring the localization of CRB requirements for sensing users and the requirements of static users through resource allocation. The macro BS has a center frequency of 2.4 GHz, a transmit power of 46 dBm, and a bandwidth of 100 MHz. The micro BS has a center frequency of 24 GHz, a transmit power of 40 dBm, and a bandwidth of 400 MHz. The noise spectrum density is −174 dBm/Hz.

As shown in Fig. 4b, a decrease in the sensing CRB results in a corresponding decrease in the communication rate, illustrating the trade-off between communication and sensing. The performance of sensing and communication in a single-BS setup degrades as the mobile user moves farther from the macro BS. If we assume a minimum communication rate requirement of 30 Mb/s, this requirement is no longer met when the user is 60 meters away from the macro BS. However, when the MNMB-ISAC network is deployed, the user enters the coverage area of the micro BS at 60 meters, and the micro BS responds to the cooperation service request. After cooperation is triggered, the performance of both sensing and communication improves significantly. Due to the lower path loss of the macro BS link, its contribution to the user's SNR is substantial. As the user continues to move, the performance decreases

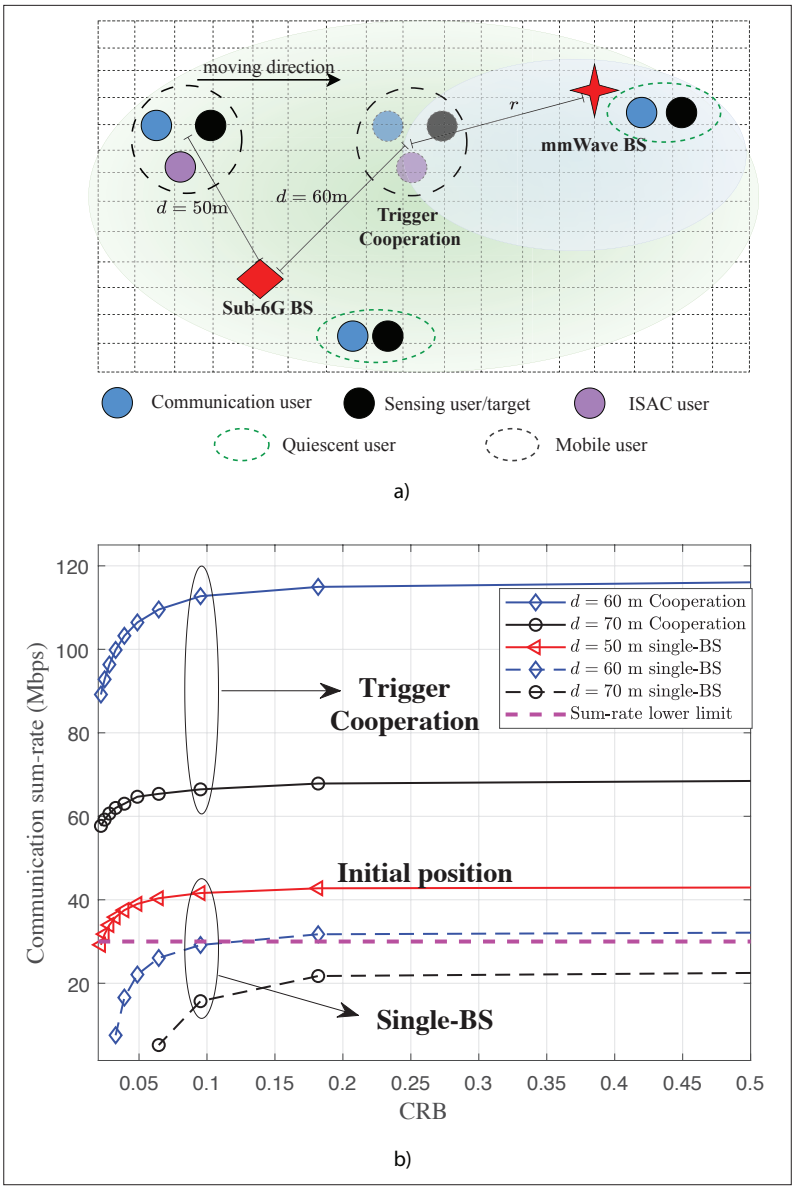


FIGURE 4. a) Schematic diagram of the simulation scenario; b) Localization CRB versus communication sum-rate.

but still meets the necessary requirements. This scenario considers cooperation between two BSs; with additional BSs, such as roadside units, the performance for mobile users would be even more stable. Figure 5 shows the resource allocation when the CRB is 0.01. In Fig .5, the micro BS allocates excess bandwidth and power resources to the users requesting cooperation after meeting the needs of its static users. Thanks to the large bandwidth of the millimeter-wave frequency band, mobile users can achieve higher sensing localization accuracy and higher communication rates.

Open Issues and Challenges

ISAC Network Architecture

Within the context of current advancements in communication technologies, an inevitable trend in the new generation of wireless networks is the implementation of the multi-band integration strategy. The crux of this strategy lies in dynamically complementing different frequency bands through

[1] Due to the stringent requirements of multi-node cooperation on time, frequency, and phase alignment, calibration can be facilitated with the assistance of user terminals. Calibration reference signals can be transmitted from each transmission point using either a single antenna or via beamforming with multiple antennas. The terminal measures these reference signals and reports auxiliary information to the network, including phase differences, frequency offsets, or time delays between other cooperative nodes and the primary node (initial access node). This feedback enables the network to dynamically adjust and align these parameters, ensuring precise synchronization and coherence for coordinated tasks.

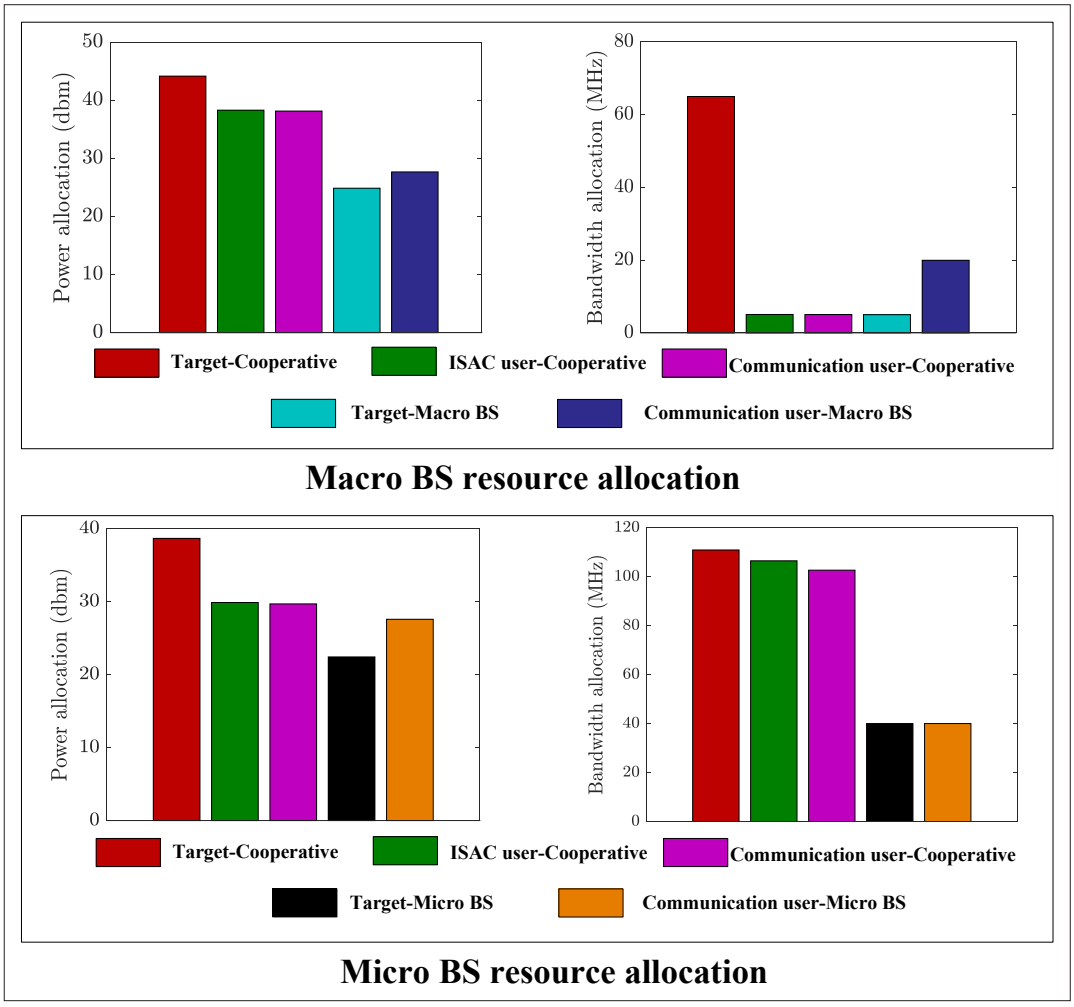


FIGURE 5. Power and bandwidth allocation results for MNMB-ISAC.

resource integration, which not only significantly enhances the overall service quality of the network but also drastically increases spectrum utilization efficiency. Nevertheless, this approach to integrated networking introduces additional challenges, particularly in terms of increased signaling overhead and network power consumption. To address these challenges effectively, designing and implementing a more flexible and efficient networking architecture is imperative. This architecture should be adaptable to diverse service requirements, thereby optimizing the allocation of network resources and substantially improving the network's service capabilities.

ISAC Interference Management and Coordination

In the evolution of ISAC technology, the imperative to augment spectral efficiency demands the investigation of resource sharing between communication and sensing systems as a pivotal research front. Concurrently, interference poses an inescapable challenge, necessitating the examination of interference cancellation and coordination tactics as critical aspects. This is especially pronounced in simultaneous and same-frequency full-duplex systems, where both self-interference and mutual interference are predominant. In order to address the strong non-linearity and memory characteristics inherent in the self-interference within the transceiver's wireless link, a polynomial model is employed to approximate the impulse response of the non-linear element. A lightweight temporal processing network is architected using recurrent neural network (RNN), long short-term memory network (LSTM), and temporal convolutional network (TCN), which possess temporal dynamics, enabling the network to accurately fit the non-linear distortion in self-interference signals with low complexity and high precision. Consequently, the impact of self-interference is eradicated within the digital signal processing unit. In order to mitigate the low utilization of time-frequency-spatial resources resulting from orthogonal schemes in shallowly coupled systems for communication and sensing signals, beamforming techniques are deployed. These techniques modulate the sensing and communication signals into distinct lobes, spatially separating the services from each other, thereby minimizing the interference effects caused by signal overlap.

ISAC Cooperation Node Selection

With respect to node cooperation, geographical proximity emerges as a pivotal factor, efficiently mitigating communication latency and energy consumption while enhancing the stability of data transmission. Channel quality, a central metric for determining cooperation efficiency, dictates that nodes with minimal signal interference and high signal strength be prioritized to ensure optimal

communication outcomes. Moreover, the energy status of nodes is crucial for maintaining system stability; hence, selecting nodes with ample energy for cooperation is imperative, alongside the implementation of effective energy management strategies for nodes with limited energy. In terms of computational capacity, cooperation nodes should possess adequate capabilities to handle both sensing and communication tasks. Nodes with higher computational performance can significantly enhance the system's data processing efficiency. In densely populated network areas, strategic placement of cooperation nodes is essential to prevent excessive resource competition and interference, ensuring efficient utilization of network resources. Concurrently, considering the current load on nodes, it is advisable to prioritize loads with lighter loads to ensure that their resources are sufficient for additional tasks. For mobile nodes, it is necessary to assess their trajectory and velocity to select those that can maintain stability during communication and sensing missions, as this will positively impact the overall system performance.

ISAC Cooperation Results Combination

A pivotal challenge in multi-node cooperation lies in effectively integrating and processing multidimensional information from various collaborating nodes to enhance the collective advantage to its fullest potential. In the multi-node cooperative data fusion phase, the system's core lies in the integrative processing of the sensing information gathered by each node, thereby enhancing the overall system performance and decision-making accuracy. This fusion process spans three dimensions: the data layer, the feature layer, and the decision layer. Data layer fusion facilitates comprehensive learning of the integrated feature information from the raw data at the data level. However, this approach necessitates a high degree of synchronization among sensing nodes, and the fusion performance may be adversely affected by the alignment of data synchronization, potentially resulting in suboptimal training outcomes. Feature layer integration, such as the preliminary extraction of features via a backbone network, followed by the amalgamation of the resultant feature maps, affords both flexibility and economy in resource utilization. However, a definitive method or protocol for this integration remains elusive; the current approaches are highly intuitive and lack a robust theoretical foundation. The decision layer fusion integrates the terminal outputs of numerous networks, rendering the system more adaptable. It allows for the modification of information integration based on varying requirements. However, this necessitates the separate training of multiple networks, which incurs a higher cost. Each perceptual node introduces various inaccuracies due to bandwidth limitations, noise, computational power, and hardware configurations during the calculation of sensing results. In order to mitigate the impact of these inaccuracies, data is meticulously filtered, retaining only that with a relatively smaller margin of error for subsequent integration.

Advanced Technologies Empowering MNMB-ISAC

Advanced physical layer technologies can significantly enhance the performance of ISAC networks. For example, non-orthogonal multiple access (NOMA) improves spectral efficiency by enabling multiple users to share the same frequency resources, while rate-splitting multiple access (RSMA) dynamically manages interference and optimizes resource allocation, supporting diverse user demands in multi-node ISAC scenarios. In addition, reconfigurable intelligent surfaces (RIS), particularly simultaneously transmitting and reflecting-RIS (STAR-RIS), act as a versatile tool for wireless channel reconfiguration. It can extend coverage, improve energy efficiency, and enhance beamforming precision. Together, these technologies address challenges such as resource constraints, interference, and coverage limitations. Exploring their integration into the MNMB-ISAC framework offers promising directions for future ISAC systems, paving the way for robust and adaptable solutions in 6G networks.

Conclusions

This article provides an integrative review of the latest advancements, challenges, and opportunities in the field of MNMB-ISAC. This study introduces an MNMB-ISAC network architecture, delving into its advantages in ISAC, multi-node cooperative sensing and communication, and multi-band cooperative sensing and communication. MNMB-ISAC networks demonstrate immense potential in resolving the scarcity of spectrum resources and supporting innovative applications. This network can provide personalized services for diverse ISAC needs, significantly enhancing network performance through cooperative resource allocation. The performance evaluations reveal the effectiveness of the MNMB-ISAC network in ensuring sensing and communication high service quality, stability of mobile services, and enhanced edge coverage in various scenarios. Furthermore, this article explores the challenges faced by MNMB-ISAC and outlines potential directions for future research. We are confident that this work will spark further innovation in the cooperative ISAC domain.

In the multi-node cooperative data fusion phase, the system's core lies in the integrative processing of the sensing information gathered by each node, thereby enhancing the overall system performance and decision-making accuracy. This fusion process spans three dimensions: the data layer, the feature layer, and the decision layer

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